

JOSHUA DIMASAKA

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RESEARCH INTERESTS

Geospatial Data Science • Earth Observation • Climate and Disaster Risk Assessment •
Earthquake Risk & Loss Modelling • Physical Risk Modelling • Exposure & Vulnerability Modelling •
Spatiotemporal Analysis • Graph Neural Networks • Probabilistic Deep Learning • Computer Vision •
Environmental Data Science • Multi-Hazard Risk Assessment • Urban Analytics

TECHNICAL SKILLS

Data & Computing: Python, R, MATLAB, PyTorch, Jupyter, Git/GitHub, SLURM/HPC, Weights & Biases
Geospatial & Earth Observation: QGIS, Google Earth Engine, GeoPandas, Rasterio, xarray, GDAL,
Remote Sensing, SAR & Optical Satellite Imagery, Spatial Disaggregation, Downscaling
Visualization & Web: Tableau, Jekyll, GitHub Pages, Mapbox, MapLibre, GeoJSON, PMTiles

EDUCATION

University of Cambridge	United Kingdom
- Doctor of Philosophy in Artificial Intelligence for the study of Environmental Risks	2023-26
Dissertation: Advanced Geospatiotemporal Modelling of Exposure and Vulnerability Using Deep Constrained Clustering and Probabilistic Temporal Graph Learning with Earth Observation Data Advisor: E. So (UCam), C. Geiß (DLR), R. Muir-Wood (Moody's RMS), F. Bendimerad (EMI)	
- Master of Research in Environmental Data Science	2022-23
Research: Exposure Assessment of Settlement-Road Systems from Climate-influenced Mass Movements Advisor: A. Marinoni (UiT), S. Selvakumaran (UCam)	
Stanford University	United States
- Master of Arts in Public Policy	2020-22
Research: Regional Earthquake Risk Assessment of the Greater Metro Manila Area, Philippines Advisor: J. Baker (Stanford)	
- Master of Science in Civil and Environmental Engineering	2019-22
Specialization: Sustainable Design and Construction (Structural Engineering) Research: Landslide, Liquefaction, and Building Damage via Causal Inference from Satellite Imagery Advisor: H. Y. Noh (Stanford), S. Xu (Johns Hopkins), D. Wald (USGS)	
- Executive Education Ignite Certificate, Entrepreneurship & Innovation	2021
- Knight-Hennessy Fellowship, King Global Leadership Program	2019-22
University of the Philippines Los Baños	Philippines
- Bachelor of Science in Civil Engineering, <i>magna cum laude</i>, college valedictorian, rank 1/341	2013-18
Tokyo Metropolitan University	Japan
- Sakura Exchange Program Certificate, Integrated River Engineering Work	2017

PROFESSIONAL APPOINTMENTS

Oct '22 - Present (4+ years)	Doctoral Researcher, UKRI CDT, University of Cambridge, United Kingdom - Improved the spatiotemporal resolution of physical exposure models of 44+ Least Developed Countries by 5x finer using geospatial ML on time-series satellite imagery and big datasets of built environment. - Established a university-industry collaboration with a nonprofit organization and a Philippine local agency for city-scale mapping of multi-hazard urban risk dynamics.
Apr - Jun '26 (¼ year)	Guest Scientist, Digital GreenTalents, German Aerospace Center, Germany - Processed 28 GB of census data with over 543 million rows of household data to produce the first nationwide maps of building typology at national, provincial, city, and neighborhood (barangay) scale

Jul - Sep '24 (¼ year)	Guest Scientist, Helmholtz AI, German Aerospace Center, Germany - Designed and implemented a deep learning framework that leverages satellite imagery and census constraints to improve spatial disaggregation for building exposure and building typology mapping.
Mar - Aug '22 (½ year)	Urban Resilience Fellow, Earthquakes and Megacities Initiative, Philippines - Developed large-scale seismic risk assessment tools covering 400,000 buildings & 3.2M population for a scenario “Big One” M7.2 earthquake.
Dec '21 - May '22 (½ year)	Public Policy Graduate Researcher, Stanford University, United States - Conducted a probabilistic seismic risk assessment of 23 cities & municipalities in Greater Metro Manila Area and examined their disaster risk management capacities with the country’s competitive index.
Mar '21 - Mar '22 (1 year)	Earthquake Risk Consultant, Stanford University (LBRE), United States - Quantified earthquake-induced financial losses and business interruption risks for 750 buildings at Stanford University under seismic scenarios up to magnitude 7.6, and conducted site visits to confirm structural vulnerabilities.
Jun - Sep '21 (¼ year)	Structures & Natural Hazards Researcher, FM Global, United States - Improved the firm’s library of earthquake design hazard and maps by more than 246%, starting with 130 and ending with 320 maps (+190) from over 130 countries.
Jun '20 - Jun '21 (1 year)	Geospatial & ML Graduate Researcher, Stanford University, United States - Implemented a Bayesian causal inference framework leveraging satellite imagery to enhance post-earthquake US Geological Survey landslide, liquefaction, & building damage models by 15x.
Jun - Sep '20 (¼ year)	Structural Retrofit Intern, South Bay Consulting Engineers, United States - Produced structural designs and resolved review comments for the retrofit of a 9,000-sq.ft. two-story home with a weak first story using a cantilevered column system.
Feb - Jun '20 (¼ year)	Housing Research Fellow, New Story Charity, United States - Evaluated 7 affordable housing material innovations and feasibility studies with 20 interdisciplinary fellows across finance, architecture, engineering, & construction.
Jul '18 - Jul '19 (1 year)	Structural Design Engineer, Arup, Philippines - Conducted post-earthquake assessment, performance- and code-based structural peer reviews for multiple 41–50 story high-rise developments.
May - Jul '17 (¼ year)	Planning & Control Engineering Intern, Ayala Group, Philippines - Built a VBA-Excel system tracking deliverables of 100+ concurrent projects.
Jun '16 - Jun '17 (1 year)	Committee Head, Engineering Congress, Philippines - Developed registration strategies that increased the national participation fivefold (300 to 1,500 attendees) for a major engineering seminar and quiz competition.

PEER-REVIEWED PUBLICATIONS

- [P-1] **Dimasaka, J.**, Bendimerad, F., Tanhueco, R. Geiß, C., and So. E. (2026). Towards Prospective Disaster Risk Management: Mapping Multi-hazard Urban Risk Dynamics Driven by Evolving Exposure and Vulnerability via Earth Observation. *Sustainable Cities and Society* (in review)
- [P-2] **Dimasaka, J.**, Geiß, C., and So. E. (2026). GraphVSSM: Graph Variational State-Space Model for Probabilistic Spatiotemporal Inference of Dynamic Exposure and Vulnerability for Regional Disaster Resilience Assessment. In *Proceedings of the Special Track on AI for Social Impact at the 40th AAAI Conference of Artificial Intelligence*.
- [P-3] **Dimasaka, J.**, Geiss, C., Muir-Wood, R., and So, E. (2026). GraphCSVAE: Graph Categorical Structured Variational Autoencoder for Spatiotemporal Auditing of Physical Vulnerability Towards Sustainable Post-Disaster Risk Reduction. *Progress in Disaster Science, Special Issue on AI, Emerging Technologies, and Immersive Solutions for Disaster & Emergency Response*.
- [P-4] **Dimasaka, J.**, Geiß, C., and So. E. (n.d.). DeepC4: Deep Conditional Census-Constrained Clustering for Large-scale Multitask Disaggregation of Urban Morphology. *Science of Remote Sensing* (in review).

- [P-5] **Dimasaka, J.**, Selvakumaran, S., and Marinoni, A. (2024). Enhancing Assessment of Direct and Indirect Exposure of Settlement-Transportation Systems to Mass Movements by Intergraph Representation Learning. *Environmental Research Letters, Focus Collection on Natural Hazards, Disasters, and Extreme Events*, 19(11), 114055.
- [P-6] Xu, S., **Dimasaka, J.**, Wald, D., and Noh, H.Y. (2022). Seismic Multi-hazard and Impact Estimation via Causal Inference from Satellite Imagery. *Nature Communications*, 13(1), 7793.

JOURNAL MANUSCRIPTS IN PREPARATION

- [M-1] Regional Dynamics of Physical Exposure & Vulnerability in Least Developed Countries.
- [M-2] From Census to Maps: A Nationwide Three-Decade Analysis of Philippine Exposure & Vulnerability, and Risk (1990–2020).

CONFERENCE PRESENTATIONS

- [C-1] Spatial Disaggregation and Temporal Projection of Building Exposure and Physical Vulnerability using Deep Constrained Clustering & Probabilistic Graph Deep Learning. *GEM Conference: From Faults to Future Scenarios, Dynamic Exposure Modelling*, Croatia, Jun 24, 2026.
- [C-2] METEOR 2.5D: An Open Geospatial Dataset of the Spatiotemporal Evolution of Physical Vulnerability in UN-recognized Least Developed Countries (as of 2020) at Five-year Intervals, 1975-2030. *EGU General Assembly 2026*, Austria, May 5, 2026.
- [C-3] A Data-Driven Probabilistic Approach to Regional Dynamics of Building Exposure and Physical Vulnerability Towards Global Disaster Risk Quantification Audit. *AGU Annual Meeting 2025*, United States, Dec 16, 2025.
- [C-4] Modernizing Quezon City's Building Risk Data with AI and Earth Observation. *20th Assoc. of Pacific Rim Universities - Multi-Hazards Conference & Symposium*, Philippines, Nov 28, 2025.
- [C-5] Spatial Disaggregation of Rwandan Building Exposure and Vulnerability via Weakly Supervised Conditional Census-Constrained Clustering (C4) using Earth Observation Data. *AGU Annual Meeting 2024*, United States, Dec 10, 2024.
- [C-6] Global Mapping of Exposure & Physical Vulnerability Dynamics in Least Developed Countries using Remote Sensing & Machine Learning. *12th Int'l Conference on Learning Representation (ICLR), 2nd Machine Learning for Remote Sensing Workshop*, Austria, May 11, 2024.
- [C-7] Near-real-time Country-wide Estimation of Susceptibility and Settlement Exposure from Norwegian Mass Movements via Inter-graph Representation Learning. *AGU Annual Meeting 2023*, United States, Dec 14, 2023.
- [C-8] Improving Post-earthquake Disaster Response using Bayesian-Updated Ground Failure Models with Satellite Imagery. *20th Assoc. of Structural Engineers of the Philippines (ASEP) International Convention*, Philippines, May 21, 2021.

INVITED TALKS

- [I-1] Near-real-time Country-wide Estimation of Susceptibility and Settlement Exposure from Norwegian Mass Movements via Intergraph Representation Learning. *Innovations Group WG5 of LandAware, an int'l network on early warning systems*, Online, Jun 25, 2024.
- [I-2] Computing in Civil Eng'g. *University of the Philippines Los Baños*, PH, Sep 29, 2023.
- [I-3] Mapping Global Disaster Risk using AI and Earth Observation Data: Examples from Climate-Induced Mass Movements and Seismic Multi-Hazard Impact Assessment. *Moody's Risk Management Solutions – Model Development Team*, London, UK, Jun 21, 2023.

TECHNICAL REPORTS

- [T-1] Tanhueco, R.M., Bendimerad, F., Khazai, B., Tibig, L., **Dimasaka, J.**, and Luzon, P.K. (2022). Climate and Disaster Risk Assessment Report for Quezon City: Climate Change, Earthquake, Flood, and Landslide Hazards, including Identification of Hotspot Barangays. Quezon City Government and Earthquakes & Megacities Initiative.
- [T-2] Bendimerad, F., Canney, N., and **Dimasaka, J.** (2021). Seismic Risk & Loss Estimation of Stanford University Buildings.

OPEN-SOURCE DATA INITIATIVES

- [O-1] METEOR 2.5D: An Open Geospatial Dataset of Spatiotemporal Evolution of Physical Vulnerability in UN-recognized Least Developed Countries (as of 2020) at Five-year Intervals, 1975-2030. doi.org/10.5281/zenodo.16608380 | doi.org/10.5281/zenodo.16695059
- [O-2] A Local-Scale Dataset of Annual Spatiotemporal Maps of Physical Vulnerability in the Cyclone-Impacted Coastal Khurushkul Community (Bangladesh) and Mudslide-Affected Freetown (Sierra Leone) (2016–2023). doi.org/10.5281/zenodo.16656471
- [O-3] A City-Scale Dataset of Annual Spatiotemporal Maps of Building Exposure and Physical Vulnerability in Quezon City, Philippines (2016–2030). doi.org/10.5281/zenodo.16655873
- [O-4] OpenSendaiBench: A Benchmark Dataset of Building Exposure and Vulnerability Dynamics for EO-based Auditing of Global Disaster Risk. doi.org/10.5281/zenodo.10840484

TEACHING

- 2025 Presenter, Advanced Animations for Scientific Oral Presentation, *University of Cambridge*
- 2023 Demonstrator, Geospatial Data Visualization (Basics of GIS), *University of Cambridge*

VOLUNTEERING

- 2026-27 Session Convener, *European Geosciences Union 2027, Exposure modeling for natural hazards*
- 2026 Reviewer, *International Journal of Digital Earth*
- Reviewer, *4th ML for Remote Sensing Workshop, ICLR*
- 2025 Reviewer, *AAAI Special Track on AI for Social Impact & AI for Urban Planning Workshop*
- Reviewer, *Updated UNDRR-International Science Council Hazard Information Profiles*
- 2024 Reviewer, *American Geophysical Union Annual Meeting 2024, OSPA*
- Reviewer, *GeoAI for a Sustainable Future in Africa Workshop*
- Reviewer, *Cambridge Journal of Science and Policy*
- Exhibitor, *Cambridge Festival*
- 2023-25 Admission Ambassador for Scholar Q&A, *Stanford Knight-Hennessy Scholars Program*
- 2023 Reviewer, *American Geophysical Union Annual Meeting 2023, OSPA*
- Participant, *Joint Workshop of Disaster Research Group & UK Alliance for Disaster Research*
- 2022 Reviewer, *Intersect: The Stanford Journal of Science, Technology, and Society*
- 2020 Fellow, *New Story Charity (Affordable Housing Solutions)*
- 2017 Lead Organizer, *12th Engineering Congress (Philippine, National)*
- 2014-22 Lead Coordinator and Interviewer, *Ramar Scholarship Foundation for Low-Income Students*

RECOGNITIONS

- 2026 EGU Outstanding Student & PhD candidate Presentation, *European Geoscience Union*
- 2025 Digital GreenTalents, *Federal Ministry of Research, Technology & Space, DE* € 5,850
- 2025 Best Overall Paper Conference, *8th Int'l Disaster & Risk Conference, CY*
- 2024 Helmholtz Visiting Researcher Grant, *Helmholtz Association, DE* € 13,040
- 2023 AGU Outstanding Student Presentation Award (OSPA), *American Geophysical Union*
- 2022-26 UKRI CDT Studentship, *Eng'g & Physical Sciences Research Council, UK* £ 106,000
- Departmental Award, *Earth Sciences, University of Cambridge, UK* £ 106,000
- 2019-22 Knight-Hennessy Graduate Fellowship, *Stanford University, US* \$ 240,000
- 2018 Int'l Publication Award, *University of the Philippines, PH*
- 2018 BPI-DOST Science Award, *BPI Foundation and Dept of Science & Tech, PH*
- 2017 Best Scientific Oral Presentation, *ASEAN-CAFMN, PH*
- 2017 3rd Place, Most Outstanding Civil Eng'g Student National Award, *PICE, PH*
- 2013-17 DOST RA 7687 & RSFI Undergraduate Scholarships, *PH* ₱ 547,000

REFERENCES

Emily So MEng PhD CEng FICE
Director, Cambridge University Centre
for Risk in the Built Environment (CURBE),
United Kingdom

Fouad Bendimerad PhD
Chairman,
Earthquakes & Megacities Initiative,
Philippines

Susu Xu PhD
Assistant Professor, Department of
Civil and Systems Engineering,
Johns Hopkins University,
United States

Marish Madlangbayan PhD
Former Head and Professor,
Department of Civil Engineering,
University of the Philippines Los Baños,
Philippines

Christian Geiß PhD
Head, Georisks, German Aerospace Center
and University of Bonn,
Germany

David Wald PhD
Seismologist,
U.S. Geological Survey (USGS),
United States

Hae Young Noh PhD
Associate Professor, Department of
Civil and Environmental Engineering,
Stanford University,
United States

Alvin Karlo Tapia DSc
Professor, Institute of Physics,
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